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A

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Abstract

- This project focuses on the design and simulation of a campus network using Cisco Packet Tracer. The network represents a real-world college environment consisting of multiple departments such as Library, Computer Lab 1, Computer Lab 2, Admin Department, and Electronics Department. Each department contains multiple computers connected through switches, and routers are used to interconnect different departmental networks.
- The primary objective of the project is to establish efficient and reliable communication between different departments using the Open Shortest Path First (OSPF) routing protocol. OSPF is a dynamic routing protocol that enables routers to automatically exchange routing information and determine the shortest path for data transmission using link-state algorithms.
- The network follows a hierarchical design where star topology is implemented within each department and tree topology is used for overall network structure. Proper IP addressing and subnetting techniques are applied to divide the network into manageable subnets. Testing and verification are carried out using commands such as ping, tracert, and ipconfig to ensure network connectivity.
- This project demonstrates practical knowledge of networking concepts and simulates real-world campus network infrastructure

1. Introduction

- A computer network is a collection of interconnected devices such as computers, routers, switches, and servers that communicate with each other to share data and resources. Networking has become an essential part of modern life, especially in organizations like colleges, hospitals, and companies where communication and data sharing are critical.
- In a college campus, different departments such as administration, library, laboratories, and academic sections require a reliable network to share information efficiently. A well-designed campus network ensures smooth communication, resource sharing, and centralized management.
- This project aims to design and simulate a campus network using Cisco Packet Tracer. The network includes multiple departments, each with its own set of computers connected through switches. Routers are used to connect these departmental networks, enabling communication between them.
- The OSPF routing protocol is used in this project to allow dynamic routing between routers. OSPF is preferred because it provides fast convergence, efficient routing, and scalability. The project helps in understanding real-world networking concepts such as topology design, routing protocols, and network configuration.

2. Problem Statement

- In a campus environment, multiple departments need to communicate with each other efficiently. However, without proper network design and routing mechanisms, communication between different departments becomes difficult and inefficient.
- The problem is to design and simulate a campus network using Cisco Packet Tracer that ensures proper connectivity between all departments. The network should use appropriate IP addressing, subnetting, and dynamic routing techniques to enable efficient data transmission.
- The solution should implement the OSPF routing protocol to allow automatic route discovery and ensure reliable communication between different networks within the campus.

3. Objectives

The main objectives of this project are:

- To design a structured and scalable campus network
- To implement dynamic routing using OSPF protocol
- To apply proper IP addressing and subnetting techniques
- To ensure communication between different departments
- To simulate the network using Cisco Packet Tracer
- To test and verify network performance using networking commands
- To understand real-world network design and implementation

4. Literature / Theoretical Background

4.1 OSI Model

➤ The OSI (Open Systems Interconnection) model is a conceptual framework used to understand network communication. It consists of seven layers:

1. Physical Layer – Deals with transmission of raw bits
2. Data Link Layer – Handles error detection and framing
3. Network Layer – Responsible for routing and addressing
4. Transport Layer – Ensures reliable data delivery
5. Session Layer – Manages sessions between devices
6. Presentation Layer – Handles data formatting and encryption
7. Application Layer – Provides services to end users

4.2 TCP/IP Model

➤ The TCP/IP model is widely used in real-world networking and consists of four layers:

- Network Access Layer
- Internet Layer
- Transport Layer
- Application Layer

➤ It is simpler and more practical compared to the OSI model.

4.3 Routing Protocols

➤ Routing protocols are used to determine the best path for data transmission. They are classified into:

- Static Routing
- Dynamic Routing

➤ Dynamic routing protocols include:

- RIP (Routing Information Protocol)
- OSPF (Open Shortest Path First)
- EIGRP

4.4 OSPF Routing Protocol

- OSPF is a link-state routing protocol that uses Dijkstra's algorithm to calculate the shortest path. It maintains a database of network topology and updates routing tables dynamically.

Advantages of OSPF:

- Fast convergence
- Scalable
- Supports large networks
- Efficient routing

4.5 Network Topology

- Network topology refers to the arrangement of devices in a network.

Types used in this project:

- **Star Topology** – Used within departments
- **Mash Topology** – Used for overall campus network
- **Hybrid Topology** – Combination of multiple topologies

5. Proposed System / Network Design

5.1 Scenario Description

- The proposed system represents a college campus network infrastructure designed to enable efficient communication and resource sharing among various departments. In a real-world educational institution, multiple departments operate simultaneously and require a reliable and scalable network to exchange data, access shared resources, and maintain centralized control.
- The network designed in this project includes the following departments:
 - **Library**
 - **Computer Laboratory 1**
 - **Computer Laboratory 2**
 - **Administrative Department**
 - **Electronics Department**
- Each department functions as an independent unit with its own set of computers and local network. However, to ensure inter-departmental communication, all departments are interconnected through routers and a centralized core network.
- Within each department, multiple PCs are connected to a switch, forming a local area network (LAN). The switch acts as a central device that facilitates communication between PCs within the department. Each departmental switch is then connected to a router, which serves as a gateway for communication with other departments.
- All departmental routers are further connected to a core network, which acts as the backbone of the entire campus network. This hierarchical structure ensures organized data flow, scalability, and ease of management.
- To enable efficient communication between different networks, the OSPF (Open Shortest Path First) routing protocol is implemented. OSPF dynamically exchanges routing information between routers and selects the shortest path for data transmission, ensuring optimal performance and reliability.
- This design reflects a real-world campus network model, providing high scalability, flexibility, and efficient communication.

Function:

- Interconnect departmental networks
- Apply routing and policies
- Act as a bridge between access and core layers

3. Core Layer

- This layer consists of the core router (or multilayer switch)
- All departmental routers are connected to this central device

Function:

- High-speed data transfer
- Backbone of the network
- Ensures reliable communication between departments

Routing Protocol Used

- The network uses OSPF (Open Shortest Path First)
- All routers are configured in Area 0
- OSPF dynamically updates routing tables and selects the best path

Topology Type

- **Star Topology** → within each department
- **Mash (Hierarchical) Topology** → overall network
- Combined → **Hybrid Topology**

5.3 Network Components

- The network consists of various hardware and logical components, each playing a specific role in ensuring proper communication.

1. Routers

- Routers are the most important devices in the network as they connect different networks and enable communication between them.

Functions:

- Perform routing between different subnets
- Maintain routing tables
- Run OSPF protocol for dynamic routing
- Forward packets based on destination IP

In this project:

- Each department has one router
- All routers are connected to the core network

2. Switches

- Switches are used to connect multiple devices within a local network.

Functions:

- Forward data based on MAC address
- Provide communication within a LAN
- Reduce network collisions

In this project:

- One switch is used in each department
- Connects multiple PCs

3. PCs (End Devices)

- PCs are end-user devices used to access the network.

Functions:

- Send and receive data
- Access shared resources
- Communicate with other departments

In this project:

- Multiple PCs are used in each department
- Configured with IP address, subnet mask, and gateway

4. Cables

- Cables are used to physically connect devices in the network.

Types Used:

- **Straight-through cable** → PC to switch
- **Straight-through cable** → switch to router
- **Serial or Ethernet cable** → router to router

Function:

- Enable physical communication between devices

5. Core Network Device

- The core device (router or multilayer switch) acts as the backbone of the network.

Functions:

- Connect all departmental routers
- Handle high-speed data transfer
- Ensure network stability

6. IP Addressing & Subnetting Table

- addressing is a fundamental concept in computer networking used to uniquely identify each device connected to a network. Every device such as a computer, router, or switch must have a unique IP address to communicate with other devices.
- In this project, IPv4 addressing is used along with subnetting to divide the network into smaller and manageable parts.

6.1 IP Addressing

- An IP address is a 32-bit numerical label assigned to each device in a network. It consists of two parts:
 - **Network ID** → Identifies the network
 - **Host ID** → Identifies the device within the network

The screenshot shows a window titled "PC6(2)" with a tabbed interface. The "Desktop" tab is selected, displaying the "IP Configuration" window for the "FastEthernet0" interface. The "IP Configuration" section has two radio buttons: "DHCP" (unselected) and "Static" (selected). Below these are five text input fields: "IPv4 Address" (192.168.30.16), "Subnet Mask" (255.255.255.0), "Default Gateway" (192.168.30.1), and "DNS Server" (0.0.0.0). The "IPv6 Configuration" section also has two radio buttons: "Automatic" (unselected) and "Static" (selected). Below these are four text input fields: "IPv6 Address" (empty), "Link Local Address" (FE80::260:2FFF:FE4B:2E00), "Default Gateway" (empty), and "DNS Server" (empty). The "802.1X" section has a checkbox "Use 802.1X Security" (unchecked), a dropdown menu "Authentication" (MD5), and two text input fields for "Username" and "Password" (both empty).

Figure 2: IP Configuration of a PC

6.2 Subnetting

- Subnetting is the process of dividing a large network into smaller subnetworks (subnets). It improves:
 - Network performance
 - Security
 - Manageability
- In this project, each department is assigned a separate subnet, ensuring that traffic is organized and controlled.

6.3 IP Addressing Scheme Used

- The network uses Class C private IP addresses (192.168.x.x range), which are suitable for small to medium-sized networks like a campus.
- Each department is assigned a /24 subnet (255.255.255.0), which provides:
 - 1 Network address
 - 254 usable host addresses
 - 1 Broadcast address

6.4 Department-wise IP Allocation Table

Department	IP Address	Subnet Mask	Gateway
Library	192.168.10.0	255.255.255.0	192.168.10.1
Lab 1	192.168.20.0	255.255.255.0	192.168.20.1
Lab 2	192.168.30.0	255.255.255.0	192.168.30.1
Admin	192.168.40.0	255.255.255.0	192.168.40.1
Electronics	192.168.50.0	255.255.255.0	192.168.50.1

6.5 Why Separate Subnets Are Used

- Each department is placed in a separate subnet because:
 - Reduces network congestion
 - Improves security
 - Makes troubleshooting easier
 - Allows efficient routing using OSPF

6.6 Role of Router in Subnetting

- Routers are used to connect different subnets.

In this project:

- Each department has its own subnet
- Routers connect these subnets
- OSPF helps routers exchange routes

7. Implementation & Configuration

➤ Router Configuration

- Routers are configured using OSPF protocol.

```
R4_CompLab1>en
R4_CompLab1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4_CompLab1(config)#router ospf 1
R4_CompLab1(config-router)#network 192.168.10.0 0.0.0.255 area 0
R4_CompLab1(config-router)#
```

☐ Top

Figure 3: OSPF Configuration on Router

```
R5_CompLab2>en
R5_CompLab2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0/0
L       10.0.0.14/32 is directly connected, GigabitEthernet0/0/0
O       192.168.10.0/24 [110/2] via 10.0.0.11, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
O       192.168.20.0/24 [110/2] via 10.0.0.12, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
O       192.168.30.0/24 [110/2] via 10.0.0.13, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0/1
O       192.168.50.0/24 [110/2] via 10.0.0.15, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0

R5_CompLab2#
```

Figure 4: Routing Table Showing OSPF Routes

➤ **Switch Configuration**

- Switches are used to connect PCs. Basic configuration is applied.

➤ **OSPF Configuration**

- Dynamic routing enabled
- Routers exchange routes automatically
- Routing tables updated dynamically

8. Testing & Verification

- Testing and verification are essential steps in any network design to ensure that the network is functioning correctly and all devices can communicate with each other without errors. In this project, various testing methods and commands are used to verify connectivity and performance.

8.1 Ping Test

- The ping command is used to check connectivity between two devices in the network. It sends ICMP (Internet Control Message Protocol) echo request packets to the destination and waits for a reply.

```
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 5: Successful Ping Between Different Departments

Explanation:

- If the destination device responds → network is working
- If request times out → network issue exists

Observation:

- Successful replies indicate proper routing via OSPF
- Time delay shows network performance

8.2 Tracert Command

The tracert (trace route) command is used to determine the path taken by data packets from source to destination.

```

C:\>tracert 192.168.40.1

Tracing route to 192.168.40.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.10.1
  2  0 ms      0 ms      0 ms      192.168.40.1

Trace complete.

C:\>tracert 192.168.30.1

Tracing route to 192.168.30.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.10.1
  2  0 ms      0 ms      0 ms      192.168.30.1

Trace complete.

C:\>

```

Figure 6: Trace Route Showing Packet Path

Purpose:

- Identifies intermediate routers
- Helps in troubleshooting routing issues

Observation:

- Displays each hop (router) in the network
- Confirms that OSPF is selecting correct paths

8.3 IP Configuration Verification

- The ipconfig command is used to verify the IP address configuration of PCs.

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::206:2AFF:FE34:7EDE
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>

```

Figure 7: IP Configuration Verification

Checks:

- IP Address
- Subnet Mask
- Default Gateway

Observation:

- Ensures correct IP assignment
- Helps detect configuration errors

8.4 Routing Table Verification

Purpose:

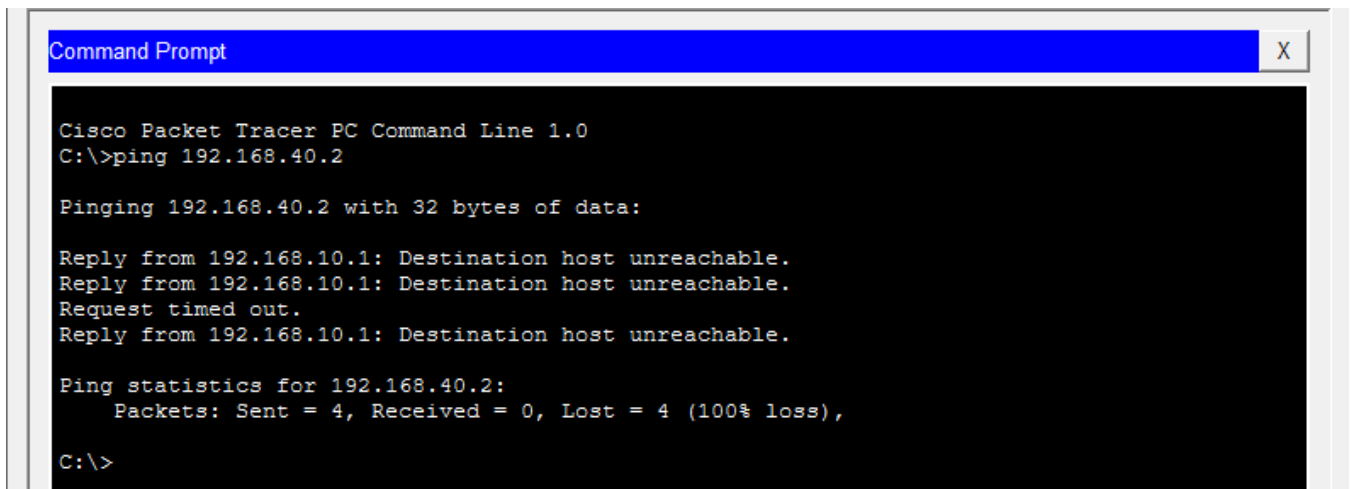
- Displays routing table of router
- Shows OSPF learned routes (marked as "O")

Observation:

- All department networks appear in routing table
- Confirms OSPF is working correctly

8.5 Fault Case

- A fault is intentionally created to test network reliability.

A screenshot of a Cisco Packet Tracer PC Command Line window. The title bar is blue and says "Command Prompt". The window has a black background with white text. The text shows a user at a C:\ prompt typing "ping 192.168.40.2". The output shows "Pinging 192.168.40.2 with 32 bytes of data:" followed by three "Reply from 192.168.10.1: Destination host unreachable. Request timed out." messages. Finally, it shows "Ping statistics for 192.168.40.2: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),". The prompt returns to "C:\>".

```
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 8: Ping Failure Due to Incorrect OSPF Configuration

Example Fault:

- OSPF configuration removed from one router

Result:

- Ping fails
- No route available

Solution:

- Reconfigure OSPF

Observation:

- Network becomes functional again

9. Results & Observations

- After implementing and testing the network, the following results were observed:
 - All departments such as Library, Admin, and Labs are successfully interconnected
 - OSPF routing protocol dynamically updates routing tables
 - Network communication between different departments is smooth and efficient
 - Minimal delay is observed during data transmission
 - Routing paths are optimized using shortest path algorithm
 - The network remains stable even when minor changes are made

Overall Result:

- The network design is successful, reliable, and efficient.

10. Advantages & Limitations

10.1 Advantages

1. Scalability

- The network can be easily expanded by adding new departments or devices without major changes.

2. Dynamic Routing

- OSPF automatically updates routing tables, reducing manual configuration.

3. Efficient Communication

- Data is transmitted using the shortest path, improving performance.

4. Reliability

- With proper configuration, the network remains stable and functional.

5. Easy Troubleshooting

- Commands like ping and tracert help identify issues quickly.

10.2 Limitations

1. Complexity

- OSPF configuration can be complex for beginners.

2. Cost

- Implementation requires routers, switches, and cables.

3. Maintenance

- Requires regular monitoring and updates.

4. Dependency on Configuration

- Incorrect configuration can cause network failure.

11. Conclusion

- The project successfully demonstrates the design and implementation of a campus network using the OSPF routing protocol. It provides a clear understanding of how different departments in a campus environment can be interconnected using routers and switches.
- The use of OSPF ensures dynamic routing, allowing routers to automatically exchange routing information and determine the most efficient path for data transmission. The hierarchical topology used in the project improves scalability and management of the network.
- Through simulation in Cisco Packet Tracer, practical knowledge of networking concepts such as IP addressing, routing, and topology design is gained. The testing and verification process confirms that the network operates efficiently and reliably.
- Overall, this project reflects real-world network design and enhances understanding of modern networking technologies.

12. References

➤ The following resources were used for understanding and implementing the project:

1. NPTEL – Computer Networks Course
2. Cisco Networking Academy (NetAcad)
3. “Computer Networking” by Kurose and Ross
4. TutorialsPoint – Networking Tutorials
5. GeeksforGeeks – Networking Concepts
6. Cisco Official Documentation
7. Online research articles and technical blogs